Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
abgabdullina@abagbdullina:~$ ls /etc/ > file.txt
abgabdullina@abagbdullina:~$ ls >> file.txt
abgabdullina@abagbdullina:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
cifs-utils
cockpit
```

Рисунок 1: Запись в файл

Работа фильтра

```
abgabdullina@abagbdullina:~$
abgabdullina@abagbdullina:~$ grep .conf file.txt > conf.txt
abgabdullina@abagbdullina:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
makedumpfile.conf.sample
man-db.conf
```

Рисунок 2: Поиск расширения .conf

Команда поиска

```
/home/abgabdullina/.npm/_cacache/content-v2/sha512/c3
/home/abgabdullina/.npm/_cacache/index-v5/1a/ce
/home/abgabdullina/.npm/_cacache/index-v5/c4
/home/abgabdullina/.npm/_cacache/index-v5/1d/f1/ce5e4168fa5f662c23cbbd558c5d82ef5e57d137e2d329fbc17126408620
/home/abgabdullina/.npm/_cacache/index-v5/68/ca
/home/abgabdullina/.npm/_cacache/index-v5/9e/e0/c3386deeef310820ca85762b73df78a4cca889774c30364b036d1f870889
/home/abgabdullina/.npm/_cacache/index-v5/c7
/home/abgabdullina/.npm/_cacache/index-v5/c7/b3/c81fcd46b7c6e984952f72b5d5823025eb825b31eb9b584107047503dd5a
/home/abgabdullina/.npm/_cacache/index-v5/f3/6c/c3986891a2eed646eaa7d914d91cffc8d6421272c959bdb5a87486645945
/home/abgabdullina/.npm/_cacache/index-v5/84/ce
/home/abgabdullina/.npm/_cacache/index-v5/09/c0
/home/abgabdullina/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3ced5cf6fc29def30220242c53
/home/abgabdullina/.npm/_cacache/index-v5/ce
/home/abgabdullina/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a58996cfad55fb2000f59e2edcc
/home/abgabdullina/.npm/_cacache/index-v5/cd
/home/abgabdullina/.npm/_cacache/index-v5/c9
/home/abgabdullina/.npm/_cacache/index-v5/c0
/home/abgabdullina/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0b53cea310cda0b203217f693
/home/abgabdullina/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86117d1350482a1d990504757
/home/abgabdullina/.npm/_cacache/index-v5/af/cf
/home/abgabdullina/.npm/_cacache/index-v5/c3
/home/abgabdullina/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca87532422f2506214f94db8861160d3
/home/abgabdullina/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837bc2c8188a44d554f3f3ba2e
/home/abgabdullina/.npm/_cacache/index-v5/ca
/home/abgabdullina/snap/hugo/common
/home/abgabdullina/snap/hugo/current
/home/abgabdullina/conf.txt
abgabdullina@abagbdullina:~$
```

Рисунок 3: Поиск файлов

Команда поиска

```
/etc/foomatic/hashes.d/hashes.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sssd': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
/etc/sysconfig/htcacheclean
```

Рисунок 4: Поиск файлов

Управление процессами

```
abgabdullina@abagbdullina:~$
abgabdullina@abagbdullina:~$ find /etc -name "h*" -print | less
find: '/etc/audit': Permission denied
find: '/etc/credstore.encrypted': Permission denied
abgabdullina@abagbdullina:~$
abgabdullina@abagbdullina:~$
abgabdullina@abagbdullina:~$ find ~ -name "log*" > logfile &
[1] 7032
abgabdullina@abagbdullina:~$
[1]+  Done                    find ~ -name "log*" > logfile
abgabdullina@abagbdullina:~$ rm logfile
abgabdullina@abagbdullina:~$
```

Рисунок 5: Фоновый запуск процесса

Управление процессами

```
abgabdullina@abagbdullina:~$
abgabdullina@abagbdullina:~$ gedit &
[1] 7070
abgabdullina@abagbdullina:~$ ps | grep gedit
   7070 pts/0    00:00:00 gedit
abgabdullina@abagbdullina:~$ kill 7070
[1]+  Terminated              gedit
abgabdullina@abagbdullina:~$
abgabdullina@abagbdullina:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

Команды df и du

```
DF(1)                                User Commands                                DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This  manual  page  documents  the GNU version of df.  df displays the amount of space available on the
       file system containing each file name argument.  If no file name is given, the space available  on  all
       currently  mounted  file systems is shown.  Space is shown in 1K blocks by default, unless the environ-
       ment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

       If an argument is the absolute file name of a device node containing a mounted file  system,  df  shows
       the  space  available  on  that  file system rather than on the file system containing the device node.
       This version of df cannot show the space available on unmounted file systems, because on most kinds  of
       systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
       Show information about the file system on which each FILE resides, or all file systems by default.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

       -B, --block-size=SIZE
              scale  sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes;
              see SIZE format below

       --direct
              show statistics for a file instead of mount point

 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

Команды df и du

```
DU(1)                                User Commands                                DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print apparent sizes rather than device usage; although the apparent size is usually smaller, it
              may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and the
              like

       -B, --block-size=SIZE
              scale  sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes;
              see SIZE format below

       -b, --bytes
              equivalent to '--apparent-size --block-size=1'

       -c, --total
              produce a grand total
 Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

Команды df и du

```
abgabdullina@abagbdullina:~$
abgabdullina@abagbdullina:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 91164764  63417044  59% /
devtmpfs         4011280        0   4011280   0% /dev
tmpfs            4035656      100   4035556   1% /dev/shm
tmpfs            1614264     1964   1612300   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journald.service
tmpfs            4035656     1640   4034016   1% /tmp
/dev/nvme0n1p3 155186176 91164764  63417044  59% /home
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop0         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop2         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop1        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolved.service
tmpfs             807128      176    806952   1% /run/user/1058
/dev/sr0         2677920  2677920         0 100% /run/media/abgabdullina/Fedora-WS-Live-43
abgabdullina@abagbdullina:~$
```

Рисунок 9: Справка по команде du

Команды df и du

```
4         ./.npm/_cacache/index-v5/52
4         ./.npm/_cacache/index-v5/85/e2
4         ./.npm/_cacache/index-v5/85
4         ./.npm/_cacache/index-v5/a7/81
4         ./.npm/_cacache/index-v5/a7
4         ./.npm/_cacache/index-v5/30/1a
4         ./.npm/_cacache/index-v5/30
424       ./.npm/_cacache/index-v5
115644    ./.npm/_cacache
115708    ./.npm
16        ./snap/hugo/25676/.config/go/telemetry/local
0         ./snap/hugo/25676/.config/go/telemetry/upload
16        ./snap/hugo/25676/.config/go/telemetry
16        ./snap/hugo/25676/.config/go
16        ./snap/hugo/25676/.config
16        ./snap/hugo/25676
0         ./snap/hugo/common
20        ./snap/hugo
20        ./snap
0         ./monthly
0         ./reports/monthly/monthly
0         ./reports/monthly
0         ./reports
4         ./ski.plases/equipment
0         ./ski.plases/plans
4         ./ski.plases
0         ./australia
0         ./play/games/play
0         ./play/games
0         ./play
1561464   .
abgabdullina@abagbdullina:~$
```

Рисунок 10: Запуск команды du

Команда поиска

```
/home/abgabdullina/.npm/_cacache/index-v5/52
/home/abgabdullina/.npm/_cacache/index-v5/52/4d
/home/abgabdullina/.npm/_cacache/index-v5/85
/home/abgabdullina/.npm/_cacache/index-v5/85/e2
/home/abgabdullina/.npm/_cacache/index-v5/a7
/home/abgabdullina/.npm/_cacache/index-v5/a7/81
/home/abgabdullina/.npm/_cacache/index-v5/30
/home/abgabdullina/.npm/_cacache/index-v5/30/1a
/home/abgabdullina/snap
/home/abgabdullina/snap/hugo
/home/abgabdullina/snap/hugo/25676
/home/abgabdullina/snap/hugo/25676/.config
/home/abgabdullina/snap/hugo/25676/.config/go
/home/abgabdullina/snap/hugo/25676/.config/go/telemetry
/home/abgabdullina/snap/hugo/25676/.config/go/telemetry/local
/home/abgabdullina/snap/hugo/25676/.config/go/telemetry/upload
/home/abgabdullina/snap/hugo/common
/home/abgabdullina/monthly
/home/abgabdullina/reports
/home/abgabdullina/reports/monthly
/home/abgabdullina/reports/monthly/monthly
/home/abgabdullina/ski.plases
/home/abgabdullina/ski.plases/equipment
/home/abgabdullina/ski.plases/plans
/home/abgabdullina/australia
/home/abgabdullina/play
/home/abgabdullina/play/games
/home/abgabdullina/play/games/play
abgabdullina@abagbdullina:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.